

Manufacturing Big Data Modeling Algorithm Based on GM (1,1) - LSTM and Its Application in Sales Forecasting

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Abstract: It is a new period for the development of automobile industry, the economic situation is complex and changing, and the policies of automobile industry are frequently issued, so accurate prediction of automobile sales is extremely important and practical for both government and enterprises. In this paper, the GM(1,1) model and the long short-term memory (LSTM) neural network model are combined and optimized, and the sales of a brand of cars from January 2019 to September 2022 are used as sample data, and the car sales in the next three months are predicted by two single models and linear combination forecasting models, respectively. The experimental results show that the linear combined forecasting model outperforms the other two single models in terms of forecasting results and has better resistance to the interference of external factors.

Key Words: Big Data Modeling, Sales Forecasting, Manufacturing Enterprise

1 Introduction

In the rapidly developing world of information technology, smart manufacturing has become a trend. In addition to maintaining the originality of manufacturing enterprises, upgrading and transformation will certainly use big data, Internet of things and other technologies. The key application of big data in manufacturing industry covers all stages of product life cycle such as counting, manufacturing, service, market and reuse, and each stage will have corresponding data. The data converged in the whole cycle are both numerous and complex. But with the support of big data, the manufacturing industry is able to move forward in a better way. However, manufacturing big data has the characteristics of multi-channel, cross-scale, high throughput, strong correlation, and heavy machinery, which makes it difficult to model manufacturing big data. Since machine learning can effectively deal with the above problems and help us mine useful information from massive data to build business models, it has been widely used in the field of data mining and modeling. In [1], Mohammadi, P. describes the challenges faced by modern manufacturing systems, such as increasingly complex, dynamic, high-dimensional and chaotic structures, and argues that machine learning techniques are effective candidates to overcome these challenges. Therefore, the use of machine learning techniques for modeling big data in manufacturing is an inevitable trend for the future development of manufacturing and is beneficial for companies to create greater value.

Applications of machine learning in manufacturing include predicting product quality, fault identification and diagnosis, and sales forecasting, etc. For example, Parsa Mo-

hammad uses a support vector machine classification algorithm to predict the product quality of burned balls based on data collected during the manufacture of wear-resistant materials[2]. In [3], Ciflikli proposed the C4.5 decision tree model for quickly identifying the causes of faults in the blanket manufacturing process, enabling those involved to quickly and accurately resolve faults and greatly improve blanket production efficiency. In [4], M. proposed a probabilistic estimation model generalized by an information network algorithm and demonstrated that it can be successfully used to determine the optimal process settings for complex manufacturing processes. In [5], Sohyung Cho proposed an intelligent tool breakage detection system using a support vector machine (SVM) learning algorithm to provide the ability to identify process anomalies and initiate corrective actions during the manufacturing process with good results. In [6], Chen built a quality predictor for analyzing the relationship between manufacturing process parameter settings and final product quality in plasma-enhanced chemical vapor deposition (PECVD) for semiconductor manufacturing by applying the back propagation neural network (BPNN) algorithm and Taguchi's method.

The limitation of the above study is that most of the existing manufacturing forecasting models have been built using a single algorithm or its improved version. Nevertheless, due to the limitations of using a single forecasting algorithm, the forecasting accuracy may not be satisfactory [7]. Therefore, a combinatorial model built by several algorithms has been proposed. In 1969, Bates conducted the first systematic study of combined forecasting, and the results attracted the attention of scholars in the field of forecasting [8]. In [9], Tian proposed a new decomposition-based combined wind speed prediction model, and the performance of the developed forecast model was verified by using two actually collected ultra-short-term wind speed and short-term wind speed datasets. In [10], Xiao integrates the K-nearest neighbor classification algorithm and the logistic regression algorithm in linear parallel to obtain a combined K-nearest neighbor-logistic regression model (KNN-LR), and uses the combined model to predict whether the product can com-

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plete the design task within the specified time. The experimental results show that the combined model has better performance compared with the model built by a single algorithm. Zheng believes that separate methods cannot build realistic prediction models and future prospects should focus on combined prediction methods[11]. The combination of nonlinear models such as gray models and neural networks shows good performance on nonlinear problems and has great potential in the field of prediction.

In summary, the combined forecasting model is one of the inevitable trends for future development. Combined forecasting models are relatively flexible and can reduce the errors caused by a single forecasting model to a certain extent. In view of this, this paper takes the monthly sales of a brand of cars from 2019 to 2022 as the research object, combines and optimizes the GM(1,1) model and the LSTM model to construct a linear combined forecasting model to forecast the sales from October to December 2022 and verify the accuracy of the combined model. The rest of the paper is organized as follows. In Chapter 2, we present the research methodology, including the two sub-algorithms used to construct the combined model and the construction method of the combined model. Chapter 3 verifies experimentally the performance of the combined model and the model constructed by a single algorithm in terms of model evaluation metrics. Finally, the paper is summarized in Chapter 4.

2 Research Method

2.1 GM(1,1) Model

Gray prediction technique is one of the important branches of gray system theory. The gray prediction GM(1,1) model is widely used in various fields because it requires less data for modeling, is computationally simple, and can better reveal the general laws in uncertain systems. The modeling process of GM(1,1) model is as follows.

We first write down the raw data as:

$$X^{(0)} = \{x^{(0)}(1), x^{(0)}(2), \dots, x^{(0)}(n)\} \quad (1)$$

where $X^{(0)}$ denotes the primitive data set and $X^{(0)}(n)$ denotes the n th element of the primitive data set. After the primary data set is constructed, a single accumulation of the primary data is performed to obtain the new time series $X^{(1)}$:

$$X^{(1)}(k) = \sum_{i=1}^k x^{(0)}(i) \quad (2)$$

$$X^{(1)} = \{x^{(1)}(1), x^{(1)}(2), \dots, x^{(1)}(n)\} \quad (3)$$

At the same time, a first-order differential equation is established for the time series:

$$\frac{dx^{(1)}}{dt} + \theta x^{(1)} = u \quad (4)$$

This model is denoted as GM(1,1). where θ and u are the gray parameters.

2.2 Long Short-Term Memory(LSTM) Model

Long short-term memory (LSTM) is a special kind of recurrent neural network (RNN), which is mainly designed

to solve the gradient disappearance and gradient explosion problems during the training of long sequences. They were introduced by Hochreiter & Schmidhuber in 1997, and were refined and popularized by many people in following work[12]. The general structure of the LSTM is shown in Fig. 1

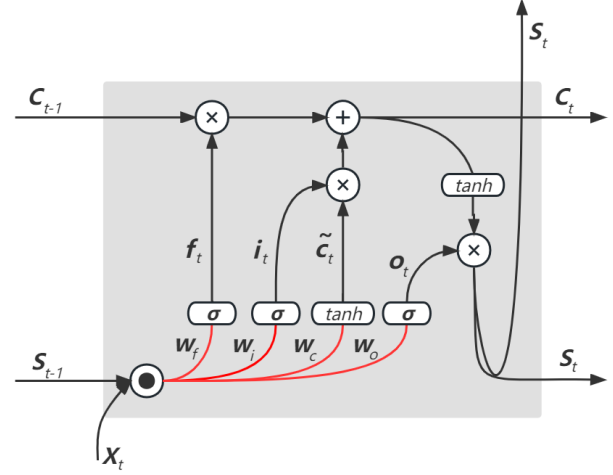


Fig. 1: Structure of LSTM

As shown in Fig. 1, there are several main stages within the LSTM. Firstly, the LSTM decide what information we're going to throw away from the cell state by a sigmoid layer called the "forget gate". It looks at S_{t-1} and X_t , and outputs a number between 0 and 1 for each number in the cell state C_{t-1} .

$$f_t = \sigma(W_f \cdot [S_{t-1}, X_t] + b_f) \quad (5)$$

The next step is to decide what new information we're going to store in the cell state. First, a sigmoid layer called the "input gate" decides which values we'll update. Next, a tanh layer creates a vector of new candidate values $\tilde{C}_t$, that could be added to the state.

$$i_t = \sigma(W_i \cdot [S_{t-1}, X_t] + b_i) \quad (6)$$

$$\tilde{C}_t = \tanh(W_C \cdot [S_{t-1}, X_t] + b_C) \quad (7)$$

Then to update the old cell state C_{t-1} into the new cell state C_t .

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (8)$$

Finally, the "output gate" decide what going to output. We run a sigmoid layer which decides what parts of the cell state we're going to output. Then, we put the cell state through tanh and multiply it by the output of the sigmoid gate, so that we can only output the parts we decided to.

$$o_t = \sigma(W_o \cdot [S_{t-1}, X_t] + b_o) \quad (9)$$

$$S_t = o_t * \tanh(C_t) \quad (10)$$

2.3 Linear Combined Model

A combined forecasting model is a type of forecasting model formed by assigning different single models with corresponding weights. The combined forecasting model is more flexible in fitting the data, which in turn helps to improve the forecasting accuracy. In this paper, a linear combination forecasting model is used to improve the reliability of the forecasting model by combining two models and giving weighting coefficients. The formula is as follows:

$$y_{0t} = w_1 y_{1t} + w_2 y_{2t} + \dots + w_n y_{nt} \quad (11)$$

where y_{0t} refers to the combined predicted value at moment t , $y_{1t}, y_{2t}, \dots, y_{nt}$ are the predicted values of different models in n at moment t ; $w_1, w_2, \dots, w_n$ are the weight coefficients of the corresponding models.

3 Experiment Analysis

3.1 Dataset

This paper selects the sales data of a brand of automobiles from January 2019 to December 2022 as the research object. The line graph is shown in Fig. 2.

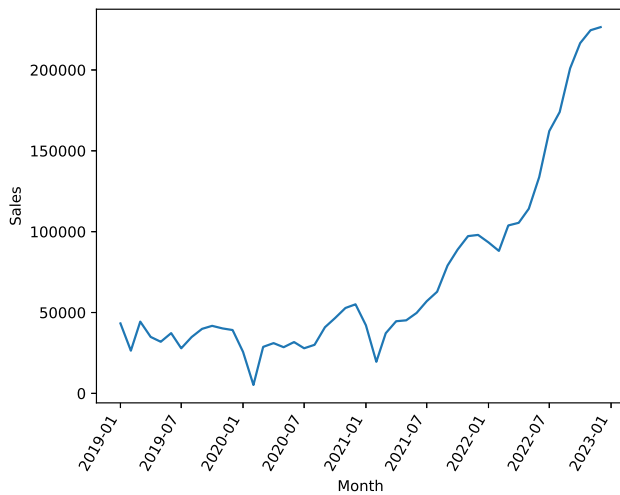


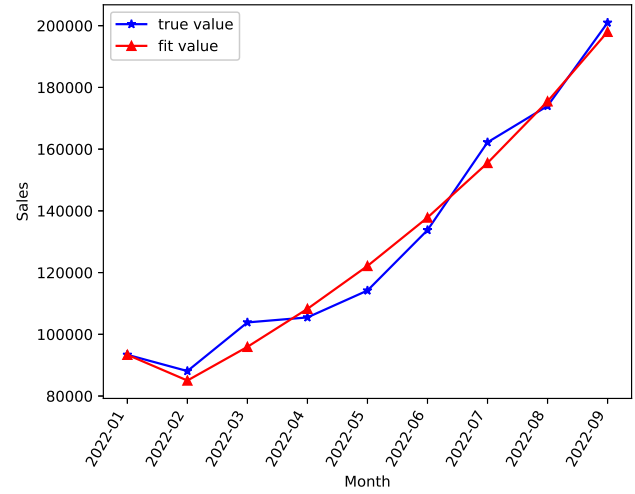
Fig. 2: Monthly sales of a brand of cars from 2019 to 2022

3.2 Establish GM(1,1) Model

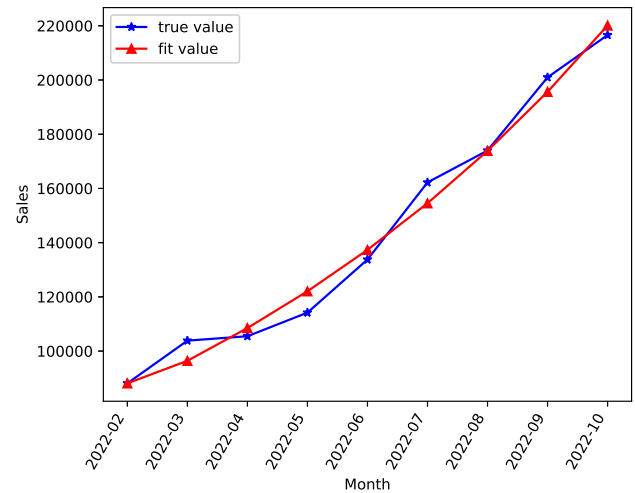
Since GM(1,1) is suitable for forecasting with a small number of samples, generally 6 to 15 samples, the data from January to September 2022 are chosen to be fitted in this paper. The model fitting results are shown in Fig. 3. From Fig. 3, it can be seen that the predicted values from October to December 2022 obtained by fitting this model are roughly close to the true values, indicating that the fitting effect is excellent.

3.3 Establish LSTM Model

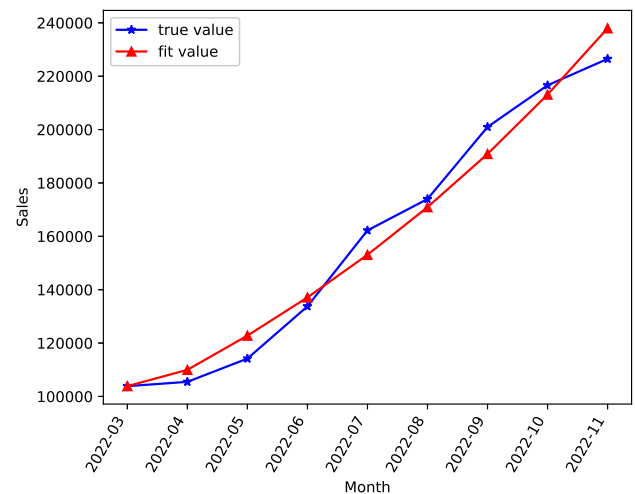
Firstly, the data were preprocessed and mapped to the interval of $[0,1]$ by the Min-Max Normalization method to make it a suitable data set for LSTM network training, and the data set was divided into training and testing sets in the ratio of 8:2; due to the small number of sample features, the single-layer LSTM structure was chosen for the neural network, and the Adam optimizer was used for training based



(a)



(b)



(c)

Fig. 3: Fitting results of GM(1,1) model for car sales

on the root mean square error (RMSE) principle. The model parameters were empirically set to a time step of 1, the number of neurons was 4, and the epoch was 1200. the model prediction results were obtained as shown in Fig. 4

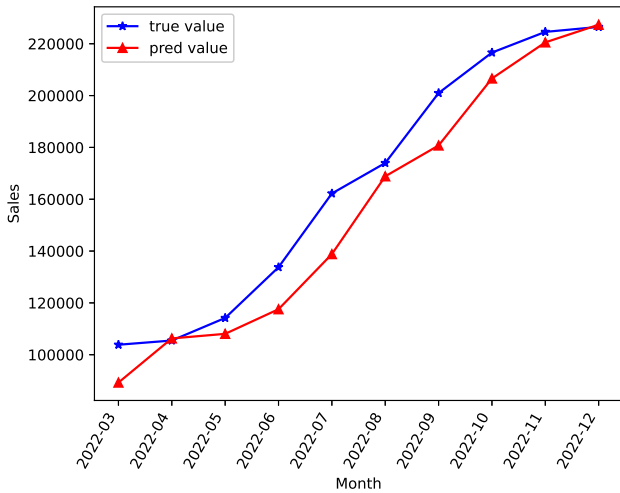


Fig. 4: Prediction results of LSTM model for car sales

3.4 Establish Linear Combined Model

Based on the prediction values obtained from the above two single models, this paper further constructs a linear combination prediction model to be combined and optimized, and the weight coefficients are obtained by the inverse of the variance method. The results of the weight coefficients are shown in Table. 1

Table 1: Weight coefficients for linear combined model

Method	Weight Coefficients
GM(1,1) Model	0.158
LSTM Model	0.842

The linear combination prediction model is obtained by bringing the weight coefficients into equation (11) as:

$$y_{0t} = 0.158y_{1t} + 0.842y_{2t} \quad (12)$$

3.5 Model Evaluation

In terms of evaluating the reliability of model prediction results, it is more reasonable to use the mean relative error (MRE) to evaluate the accuracy of the model, and generally the smaller the mean relative error, the higher the accuracy of the model is indicated. In view of this, this paper uses the above three models to forecast the auto sales data from October to December 2022 and compares the mean relative errors of different models, and the forecast results and mean relative errors are shown in Tables 2 and 3.

Table 2: Sales forecast results for a brand of cars from October to December 2022

Month	True vaule	GM(1,1)	LSTM	Combined Model
2022/10	216593	223426	206594	217400
2022/11	224576	237949	220527	224142
2022/12	226492	250564	227344	230884

Table 3: Prediction error and mean relative error of each model

Month	GM(1,1)	LSTM	Combined Model
2022/10	3.15%	4.62%	0.37%
2022/11	5.95%	1.80%	0.19%
2022/12	10.63%	0.38%	1.94%
MRE	6.58%	2.27%	0.83%

As can be seen from Table 3, the linear combination forecasting model has the smallest average relative error of 0.83%, compared with the GM(1,1) model and the LSTM model with errors of 6.58% and 2.27%, respectively, which are obviously less volatile, so the linear combination forecasting model has higher forecasting accuracy and better resistance to interference from external factors than the single forecasting model.

4 Conclusion

In this paper, firstly, the combined prediction model is obtained by combining the GM(1,1) model and the long short-term memory (LSTM) model by linear parallel integration. And the weight coefficients of the two sub-models are obtained by the inverse of variance method. Finally, by comparing the prediction results and the average relative errors of the two models built by the combined prediction model and the single algorithm, it is concluded that the combined prediction model has better prediction accuracy and wider applicability than the model built by the single algorithm.

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